# Person 1 Presentation Notes Data and Grid Foundation

Purpose: study these notes before presenting the assigned slides. Explain the ideas in your own words and use the examples when the audience is unfamiliar with AI or power grids.

## What this person must understand

This section explains the real-world problem, the two input datasets, the cleaning work, the extracted features, the country graph, and the assumptions behind the risk screen and loss estimate. The key idea is simple: the model cannot make a useful forecast until the raw rows are made consistent.

| Fact | Project value |
|---|---|
| Dynamic records | 315,598 |
| Power-flow records | 1,262,592 |
| Countries / nodes | 12 |
| Canonical links / edges | 24 |
| Time coverage | 2022–2024 hourly replay |

## Basic vocabulary

- A node is a country or grid location.
- An edge is an interconnector or power line between two nodes.
- Net flow is the signed overall flow value used by the dataset. The sign indicates direction according to the source convention.
- A feature is an input signal. The target is the future net flow the model must predict.
- A graph is a network representation that keeps countries and their connections together.

## Slide-by-slide explanation

### Slide 1  Graph-Based Predictive and Prescriptive Analytics

Purpose: Introduce the project in one sentence.

- The project combines forecasting, graph analysis, risk screening, and route testing.
- The dashboard uses a historical replay of supplied data, not live utility telemetry.

**What to say:** We turn historical cross-border power-flow records into an explainable forecast and a recommendation about which route an engineer should investigate next.

### Slide 2  The decision problem

Purpose: Explain why the grid is a network problem.

- Power can move across several connected countries at the same time.
- A stressed line can make neighboring links relevant.
- Prediction asks what may happen next; prescription asks what could be investigated.

**What to say:** Think of the grid as roads and power as traffic. If one road becomes crowded, we first predict the traffic and then test whether another connected road has space.

### Slide 3  Data becomes a model-ready window

Purpose: Explain how raw CSV records become model inputs.

- Parse timestamps and align countries by hour.
- Normalize country and interconnector aliases.
- Check duplicates, missing values, and timestamp gaps.
- Build a 24-hour input window and use the next hour as the target.

**What to say:** The model does not read a raw CSV row directly. The pipeline creates a clean tensor containing recent local behavior, network context, and graph features.

### Slide 4  Graph, risk screen, and loss assumptions

Purpose: Explain nodes, edges, risk, and the proxy physics.

- Node colors show Normal, Warning, or Critical screening states.
- Risk pressure is mainly driven by utilization, with imbalance and load residual adding context.
- Coordinates estimate line distance, but the result is not a complete AC power-flow calculation.

**What to say:** The graph tells us who is connected to whom. The risk score tells us where to look first. The loss estimate is transparent, but an engineer must replace the assumptions with verified line data before operational use.

## Equations to know

$$R = \rho L / A$$

*Resistance equals resistivity times line length divided by conductor area.*

$$P_loss = 3I^2R$$

*Estimated three-phase resistive loss from current and resistance.*

$$\text{risk score} = 75 \times \text{utilization} + 15 \times |\text{imbalance}| + 10 \times \text{residual-to-load ratio}$$

*The implementation caps the component ratios and clips the final score to 0–100.*

## Questions you should be ready for

- Why use a graph? Because neighboring countries and alternate paths affect the decision.
- Is distance exact? No. It is derived from available coordinates.
- Is the loss engineering grade? No. It is a screening proxy using assumed constants.
- Is the data live? No. The visible movement is a replay of supplied historical records.

## Handoff sentence

The data and graph are now ready. The next section explains how the project forecasts the next hour, measures performance, and tests a possible route.